

SENTRIX

Intelligent Multimodal Home Security System

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

Project Title: Sentrix — Intelligent Multimodal Home Security System

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1. Project Overview

Sentrix is an AI-based home security system that detects threats using video and audio data and assists in automated emergency response.

Sentrix is an advanced AI-powered home and premises security platform that integrates real-time multimodal analysis — computer vision, behavioural intelligence, audio anomaly detection, identity verification, and cloud-based threat detection — into a unified threat-assessment system. Unlike conventional systems that operate on simple motion or sound thresholds, Sentrix employs context-aware processing to determine whether a detected person is authorised, whether observed movement patterns suggest legitimate activity or a potential threat, and whether environmental signals indicate fire, smoke, or an armed intrusion.

The system employs a parallel multi-engine architecture in which scores from eight independent AI detection engines are aggregated by a central FusionEngine into a single Threat Confidence Index (TCI). The TCI is mapped to one of five discrete threat levels, each corresponding to a pre-defined escalating response. This design significantly reduces false alarms while ensuring that genuine high-risk events trigger an automated emergency dispatch workflow that places actionable information before the user and relevant authorities within seconds.

When a Level 4 or Level 5 event is detected, the system automatically captures a timestamped, annotated screenshot, pre-populates an emergency dispatch form with the user's registered details and AI-determined incident type, and delivers it to the user's mobile application. A single confirmation tap dispatches the complete incident package to the appropriate authority. All core AI inference is performed on-premise using edge computing, with only AES-256-GCM encrypted evidence clips transmitted externally on user command, thereby preserving household privacy and maintaining system functionality during internet outages.

2. Problem Statement

The global home security market continues to rely on outdated architectures with limited real intelligence, despite widespread AI claims. Most systems generate extremely high false alarm rates estimated at 94–98%, triggered by pets, environmental motion, or lighting changes, resulting in nearly USD 2 billion in wasted emergency resources annually in the United States [1]. At the same time, rising property crimes, such as the 2.27 lakh burglary and household theft cases reported in India in 2022 [13], combined with urban police response times of 15–45 minutes, highlight the urgency for faster and more reliable solutions. Although markets like India’s smart home security sector, valued at USD 2.1 billion in 2023 and growing at a CAGR of 22.4% [14], are expanding rapidly, existing systems still lack behavioural understanding, cannot distinguish residents from intruders, fail to maintain identity persistence across cameras, and offer little to no audio intelligence for detecting critical threats like glass breaks or distress signals. Additionally, the gap between detection and response forces users to manually assess and act under stress. Sentrrix addresses these combined global challenges by delivering a unified, AI-driven, multimodal security system deployable on consumer-grade hardware.

3. Need Analysis

Home intrusions in India result in an average police response time of 15–45 minutes in urban areas, considerably longer in suburban and rural settings. Fire response times average 8–14 minutes from point of call. In both scenarios, the operative bottleneck is the time required for the affected person to process the event, identify an appropriate authority, and communicate the situation coherently under extreme stress. The following points establish the specific need for an advanced multimodal AI security system.

1. Existing consumer security systems lack context-aware analysis. Motion-based triggers produce 94–98% false alarm rates, eroding user trust and overwhelming emergency services. A system incorporating behavioural classification and verified-quality alerting is required to restore operational value to residential security.
2. No affordable product currently offers cross-camera identity persistence. Tracking an individual's authorisation status consistently across multiple camera feeds is essential to prevent unnecessary alerts when authorised occupants move through a monitored premises.
3. Audio intelligence is absent from all consumer-tier systems. The ability to classify acoustically distinct events — gunshots, glass breaks, screams, and smoke alarms — as independent threat signals substantially enhances detection accuracy and scope.
4. The emergency response pipeline in all current systems is entirely manual. A pre-populated, one-tap dispatch mechanism that delivers incident context directly to authorities eliminates the critical delay introduced by manual communication during a live security event.
5. Enterprise-grade AI security platforms such as Verkada and Axis are priced at USD 3,000–10,000 per camera [\[2\]](#), placing professional-grade threat intelligence beyond the reach of residential users and small businesses. A system delivering equivalent analytical capability on consumer hardware addresses a significant accessibility gap in the market.

4. Literature Survey

A survey of existing work across the relevant technical domains — object detection and computer vision, person re-identification, audio event classification, IoT-integrated surveillance, and data security — provides the justification for the architectural decisions adopted in Sentries.

4.1 Object Detection and Computer Vision for Surveillance

The YOLO (You Only Look Once) family of real-time object detection architectures has established the practical benchmark for video-based threat detection. Redmon and Farhadi [3] demonstrated that single-pass convolutional detection networks can achieve competitive accuracy at inference speeds suitable for live surveillance streams. Subsequent iterations, culminating in YOLOv8 [4], have further improved small-object detection accuracy and multi-class simultaneous detection — both critical requirements for identifying persons, poses, and weapons in a single inference pass. Wang et al. [5] evaluated the performance of YOLOv8 on surveillance-specific datasets, confirming sub-50 ms detection latency on mid-range GPU hardware, which establishes its suitability as the primary vision backbone in Sentries. Standard benchmark datasets including COCO (vision) and MS-COCO variants are employed for model training and evaluation.

4.2 Person Re-Identification Across Multiple Cameras

Person re-identification (ReID) across non-overlapping cameras is a well-studied problem in computer vision. Hermans, Beyer, and Leibe [6] demonstrated that deep metric learning using triplet loss, trained to minimise intra-class embedding distances and maximise inter-class distances, outperforms classification-based approaches for cross-view person matching. Sun et al. [7] proposed the Part-based Convolutional Baseline (PCB), which partitions a person's appearance descriptor into spatial segments to improve fine-grained identity matching under viewpoint and lighting variation. These methodologies directly underpin the Sentries ReID module, which maintains persistent global identity assignments across all four camera feeds throughout a monitoring session.

4.3 Audio Event Classification and Anomaly Detection

Acoustic event detection has received growing attention as a complementary modality to vision-based surveillance. Salamon and Bello [8] demonstrated that CNN models applied to mel-spectrogram representations of audio signals achieve strong generalisation on environmental sound classification benchmarks, including the ESC-50 dataset comprising 50 categories of environmental sounds. Hershey et al. [9] introduced AudioSet, a large-scale audio dataset with 527 sound classes including gunshots, glass breaks, and screams, which constitutes the primary training resource for the Sentries audio intelligence engine. Standard datasets such as AudioSet and ESC-50 are used for model training and evaluation of the audio classification module.

4.4 IoT-Integrated Smart Surveillance Systems

Several studies have validated IoT pipeline architectures for real-time sensor data transmission to cloud dashboards and mobile applications — an architectural template directly applicable to Sentries's monitoring and dispatch pipeline. Mukundaswamy et al. [10] implemented an IoT-enabled automated monitoring system with real-time sensor data streamed to a cloud server and user notification via smartphone application, demonstrating sub-second notification latency using MQTT-based communication. Karima et al. [11] employed an NVIDIA Jetson edge processing unit for on-device image analysis, transmitting only event-triggered clips to the cloud — closely mirroring Sentries's on-premise inference and selective cloud transmission model. These works confirm that the proposed communication architecture is technically feasible and operationally validated.

4.5 Evidence Security and Data Integrity

The use of cryptographic mechanisms for evidence integrity is well-established in digital forensics. AES-256-GCM (Galois/Counter Mode) provides authenticated encryption, combining confidentiality with a cryptographic authentication tag that detects any post-collection tampering [12]. SHA-256 hashing of individual frames creates a verifiable content fingerprint such that any modification to stored evidence is cryptographically detectable.

4.6 Smart Surveillance Research in the Indian Context

Recent research from Indian institutions has advanced IoT-integrated surveillance specifically suited to India's urban infrastructure constraints. Sharma et al. [15] proposed a low-cost edge-computing surveillance framework validated on Indian urban datasets, demonstrating that

YOLOv5-based detection achieves acceptable accuracy on consumer-grade hardware under Indian lighting and density conditions directly validating Sentrix’s hardware accessibility objective. The NCRB’s annual crime statistics further contextualise the Indian deployment target, confirming that residential burglary and property crime constitute a significant and growing public safety challenge in Indian urban centres [13].

Author / Work	Year	Method	Key Contribution	Relevance to Sentrix
Redmon & Farhadi [3]	2018	YOLOv3 single-pass CNN	Real-time object detection at competitive accuracy and speed	Foundation of YOLOv8 vision backbone used in Sentrix
Wang et al. [4]	2023	YOLOv8 on surveillance datasets	Sub-50 ms detection latency; multi-class simultaneous detection	Validates YOLOv8 for live multi-camera threat detection
Hermans et al. [6]	2017	Triplet loss deep metric learning for ReID	Improved cross-view identity matching over classification baselines	Basis of cross-camera identity persistence module
Sun et al. [7]	2018	Part-based Convolutional Baseline (PCB)	Fine-grained spatial identity descriptors robust to viewpoint change	Applied to appearance embedding in Sentrix ReID engine
Salamon & Bello [8]	2017	CNN on mel-spectrograms; ESC-50	Strong generalisation on 50-class environmental sound benchmark	Basis of audio threat classifier in Sentrix
Hershey et al. [9]	2017	AudioSet — large-scale audio dataset	527-class audio taxonomy including gunshots, screams, glass-break	Primary training data source for audio intelligence engine
Mukundaswamy et al. [10]	2024	IoT sensor monitoring + cloud dashboard	Sub-second MQTT notification latency; smartphone alerts	Validates IoT pipeline architecture for Sentrix dispatch system
Karima et al. [11]	2024	Edge (Jetson) + selective cloud upload	On-device inference; event-triggered encrypted clip upload	Confirms on-premise inference + selective transmission model

Table 4.1. Comparison of AI-based surveillance and security systems.

5. Objectives

1. To develop a multimodal AI-based real-time threat detection system integrating computer vision (YOLOv8), audio event classification (CNN on Audio Set / ESC-50), and behavioural analysis to accurately identify security threats from multi-camera video and audio streams.
2. To implement a cross-camera identity tracking module using deep metric learning-based person re-identification (ReID) and face recognition, enabling persistent authorisation status assignment across all camera feeds simultaneously.
3. To design and validate a fusion-based threat scoring framework (Threat Confidence Index) that aggregates normalised outputs from all detection engines with temporal smoothing and hard override rules, mapping threat levels to a structured five-level response protocol.
4. To build an automated emergency response pipeline comprising a React Native mobile application, incident package pre-population, one-click authority dispatch, and a tamper-evident AES-256-GCM encrypted evidence chain, with performance evaluation against defined accuracy, latency, and false alarm rate metrics.

6. Methodology

The proposed system employs a parallel multi-engine architecture in which eight independent AI detection engines process continuous video and audio streams concurrently, with their outputs aggregated by a central Fusion Engine. The overall system architecture is illustrated in Figure 6.1. The stepwise methodology is described below.

1. **Data Acquisition (Input Stage):** Video and audio streams are captured from up to four IP cameras and connected microphones for downstream processing. Video is decoded at 15–30 FPS per channel and audio is sampled at 16 kHz with pre-amplification and noise gating applied.
2. **Parallel Engine Processing:** Eight AI engines execute concurrently to extract independent threat signals from the acquired streams. The Computer Vision Engine (YOLOv8) performs person detection, pose estimation, and weapon identification; the Audio Intelligence Engine classifies acoustic events using a CNN trained on MFCC spectrograms derived from AudioSet and ESC-50; the Identity Engine performs face recognition against an enrolled visitor database; the ReID Engine extracts appearance embeddings for cross-camera identity persistence; the Behaviour Engine classifies motion trajectories; and the Cloud Threat Engine queries external threat intelligence feeds.
3. **Fusion Engine and TCI Computation:** Normalised confidence scores from all engines are combined using a predefined weight vector to produce the composite Threat Confidence Index (TCI). A sliding-window temporal smoother eliminates transient false spikes, and a hard override rule maps the TCI to Level 4 or 5 immediately upon confirmed weapon or fire detection.
4. **Threat Response Execution:** The computed TCI is mapped to one of five discrete threat levels, each triggering a pre-defined escalating response ranging from event logging at Level 1 to automated authority dispatch at Level 5.
5. **Emergency Dispatch:** Upon Level 4 or 5 detection, a timestamped annotated screenshot is captured, the incident type is classified, an emergency form is pre-populated with the user's registered details, and an actionable notification is pushed to the mobile application for one-tap authority dispatch via Twilio.

6. Evidence Preservation: All Level 3–5 frames are encrypted using AES-256-GCM and stored with a SHA-256 content hash and full metadata, producing a tamper-evident evidence chain suitable for legal proceedings.

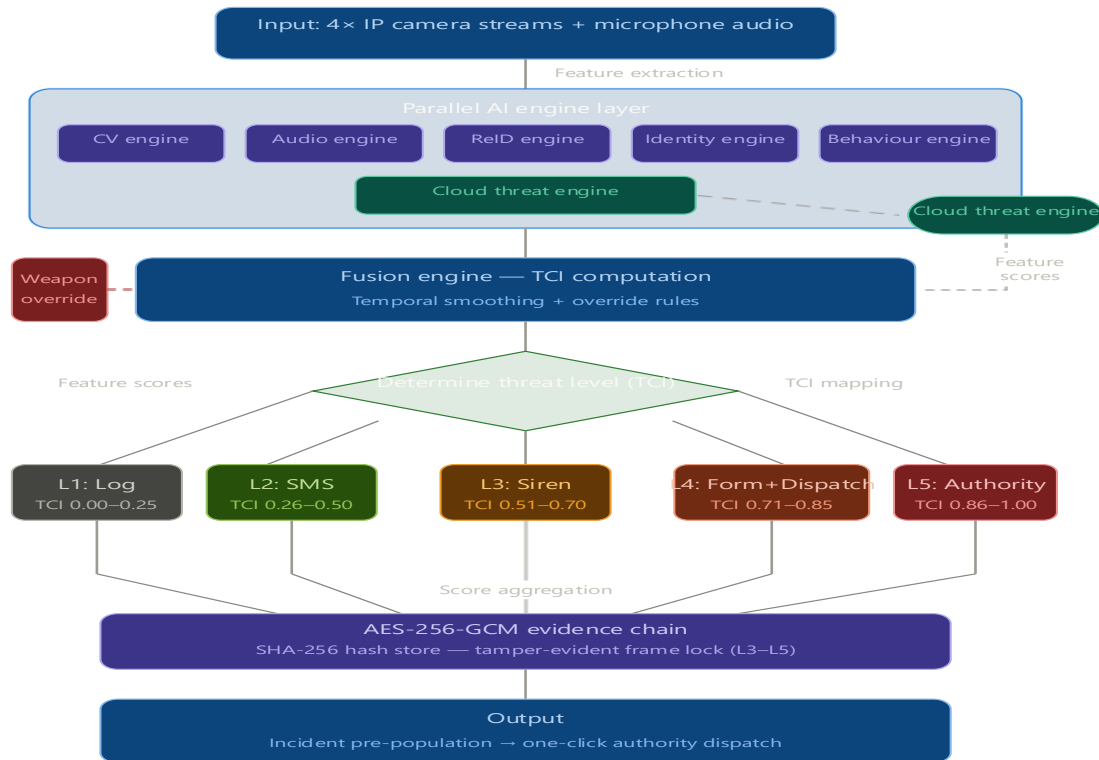


Figure 6.1. Architecture Diagram.

Level	Status	TCI Range	Trigger Condition	System Response
Level 1	Normal	0.00–0.25	No threat; routine activity.	Log only; HUD overlay on dashboard.
Level 2	Suspicious	0.26–0.50	Unusual activity or unknown person.	Snapshot saved; SMS alert with image preview sent to user.
Level 3	Elevated	0.51–0.70	Multiple risk factors converging.	SMS + local audio siren + app push notification.
Level 4	High	0.71–0.85	Confirmed threat — weapon or intrusion detected.	Auto-clip screenshot; pre-fill emergency form; one-click dispatch option.
Level 5	Critical	0.86–1.00	Weapon / fire confirmed / violent intrusion.	AES-GCM evidence lock; automated Twilio call; one-click authority dispatch.

Table 6.1. TCI five-level threat response framework.

7. Work Plan

Sr. No.	Activity	Month	July			August			September				November				December			
		Week no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Identification formulation and Planning of project	Plan																		
		Actual																		
2	CV Engine Development (YOLOv8, Pose, Weapon)	Plan																		
		Actual																		
3	Audio Intelligence Engine	Plan																		
		Actual																		
4	ReID & Identity Module	Plan																		
		Actual																		
5	FusionEngine & TCI Integration	Plan																		
		Actual																		
6	Mobile App & Emergency Dispatch	Plan																		
		Actual																		
7	System Integration & Field Testing	Plan																		
		Actual																		
8	Evaluation & Final Report	Plan																		
		Actual																		

Table 7.1. Project Work Plan

8. Project Outcomes & Individual Roles

The proposed system, Sentries, is expected to deliver the following outcomes upon successful completion:

1. Multi-Camera AI Security Platform: A functional real-time security system processing video from up to four IP cameras simultaneously on consumer-grade hardware, achieving a system throughput of ≥ 15 FPS per channel with end-to-end detection latency of < 2 seconds.
2. Audio Intelligence Engine: An audio classifier achieving $> 90\%$ classification accuracy on ESC-50 and AudioSet evaluation subsets, capable of distinguishing gunshots, screams, glass breaks, and fire alarms from ambient background noise in real time.
3. Cross-Camera Identity and ReID Module: A persistent identity tracking system achieving ReID rank-1 accuracy $\geq 85\%$ on evaluation sequences, maintaining consistent authorisation status for each tracked individual across all camera feeds.
4. FusionEngine and TCI Framework: A calibrated five-level threat scoring system with temporal smoothing and weapon/fire override, achieving a false positive rate reduction of $\geq 60\%$ compared to a motion-only detection baseline.
5. Mobile Application and Emergency Dispatch: A React Native mobile application providing real-time threat dashboard, push notifications, and one-tap authority dispatch, achieving an end-to-end dispatch latency of ≤ 10 seconds from Level 5 event detection.
6. AES-GCM Tamper-Evident Evidence Chain: An encrypted, SHA-256 hashed frame archive producing forensically admissible evidence packages automatically at Level 3 and above, with zero evidence tampering tolerance.
7. Performance Evaluation Report: A comprehensive report covering TCI precision and recall across all threat levels, audio classification accuracy, ReID performance, system latency, and false alarm rates, validated against real-world test scenarios.

Individual Roles:

1. Kartik Garg — App Development
2. Akshay Ranveer — User Interface and Documentation
3. Prashant Gagneja — Core Machine Learning Implementation
4. Harshit Mishra — Core Machine Learning Implementation
5. Mehul Perimal — Hardware Development and Integration

9. Course Subjects

1. Artificial Intelligence and Machine Learning — Applied for YOLOv8-based computer vision, CNN audio event classification, behavioural pattern analysis, and deep metric learning for person ReID.
2. Computer Networks and Internet of Things — Applied for IP camera stream ingestion, MQTT/WebSocket communication pipelines, Twilio API integration, and mobile application network architecture.
3. Operating Systems — Applied for multi-threaded parallel engine execution, process scheduling, and real-time memory management across concurrent AI inference modules.
4. Database Management Systems — Applied for storing authorised visitor profiles, system event logs, TCI history records, and encrypted evidence metadata.
5. Software Engineering — Applied for modular system architecture design, RESTful API development, test-driven development, and integration testing across eight parallel modules.
6. Cryptography and Cyber Security — Applied for AES-256-GCM implementation, SHA-256 evidence hashing, secure key management, and tamper-evident storage design.
7. Human-Computer Interaction and UI/UX Design — Applied in the design of the real-time threat dashboard and the mobile application emergency dispatch workflow.

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